Spotify Music Data Analysis

Project Documentation | PostgreSQL + Power BI

Project Overview

This project analyses Spotify Top 50 music data using PostgreSQL for SQL-based analysis and Power BI for interactive data visualisation.

■ Analyse song, artist, popularity, duration, album type, release, and chart-position data.

■ Combine SQL analysis with interactive Power BI reporting.

■ Use DAX measures to calculate KPIs and analytical metrics.

Business Requirement

The dashboard is designed to provide a consolidated view of Spotify data for analysing music performance and trends.

1. Overview Analysis

■ Monitor Total Songs, Distinct Artists, Average Popularity, and Average Song Duration.

■ Compare Explicit and Non-Explicit Songs.

■ Analyse song distribution by Album Type.

■ Track Distinct Songs released over time.

■ Analyse popularity and song trends over time.

■ Identify the Top 10 Longest Songs.

2. Artist Analysis

■ Identify the Top 10 Artists by Number of Unique Songs.

■ Compare artists by Average Popularity.

■ Identify Artists with the Most #1 Hits.

■ Review artist-level metrics including songs, #1 hits, albums, tracks, and popularity.

3. Song Analysis

■ Identify Top Songs with Multiple Artists.

■ Find the Top 10 Songs by Total Popularity.

■ Compare Explicit Songs across Album Types.

■ Explore song-level performance and popularity.

Problem Statement

The raw Spotify Top 50 dataset contains useful information, but analysing individual rows makes it difficult to quickly identify patterns and compare performance.

■ No centralised KPI monitoring — key metrics are consolidated into one dashboard.

■ Limited artist performance visibility — artist-level analysis highlights songs, popularity, and #1 hits.

■ Difficulty comparing song popularity — songs can be ranked and compared using popularity measures.

■ Limited trend visibility — time-based visuals support release and popularity analysis.

■ Explicit content analysis — explicit and non-explicit songs can be compared.

■ Raw data is difficult to interpret — interactive visuals improve exploration and communication.

SQL Analysis

PostgreSQL was used to query and analyse the Spotify dataset.

■ 1. Find all unique songs.

■ 2. Find the total number of unique artists.

■ 3. Find the total number of songs for each album type.

■ 4. Find the average popularity for each album type.

■ 5. Find the Top 10 artists with the highest number of songs.

■ 6. Find the Top 10 longest songs.

■ 7. Find the number of explicit and non-explicit songs.

■ 8. Find the number of songs released each year.

■ 9. Find the Top 10 songs by total popularity.

■ 10. Find artists with more than one song.

Power BI & DAX Analysis

The interactive Power BI report contains Overview, Artists, and Songs pages.

■ Total Songs and Distinct Songs

■ Distinct Artists

■ Average, Maximum, and Minimum Popularity

■ Average, Maximum, and Minimum Song Duration

■ Explicit and Non-Explicit Songs

■ Percentage of Explicit Songs

■ Average Chart Position and Position #1 analysis

■ Average Tracks per Album

■ Singles and Albums Count

■ Artist-level and Year-level measures

Key Insights Enabled

■ Identify artists with the highest number of unique songs.

■ Compare artist popularity and #1 chart performance.

■ Identify songs with the highest total popularity.

■ Compare popularity across album types.

■ Analyse explicit versus non-explicit content.

■ Explore music release patterns over time.

■ Identify the longest songs.

■ Identify songs featuring multiple artists.

Tools & Technologies

Tool / Technology	Purpose
PostgreSQL	Database management and SQL analysis
SQL	Data querying, aggregation, and analysis
Power BI	Interactive dashboard and visualisation
DAX	Measures and analytical calculations

Project Outcome

■ Provides an end-to-end Spotify music analytics project.

■ Demonstrates PostgreSQL, SQL, Power BI, DAX, KPI development, and data visualisation.

■ Makes song and artist performance easier to analyse and communicate.

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